Summary of Conservation Laws

Material Derivative

The *material derivative* describes the total change of a field property experienced by a moving particle. For a general property B,

$$\frac{DB}{Dt} = \frac{\partial B}{\partial t} + \boldsymbol{V} \cdot \boldsymbol{\nabla} B$$

Note: a boldface term means a vector or tensor.

Reynolds Transport Theorem

The *Reynolds transport theorem* extends the material derivative to describe the total change of an intensive property changing inside the control volume. For a general intensive property, $b = B/m$,

$$\frac{D}{Dt}\int_{\mathbb{V}} \rho b \, d\mathbb{V} = \frac{\partial}{\partial t}\int_{\mathbb{V}} \rho b \, d\mathbb{V} + \int_{A} \rho b \boldsymbol{V} \cdot \hat{\boldsymbol{n}} \, dA$$

The first term describes total change of B for the system, the second term describes time changes, and the third describes flows in/out of the system.

Divergence Theorem

The *divergence theorem* allows us to interchange area and volume integrals. Some examples,

$$\int_{A} \rho b \boldsymbol{V} \cdot \hat{\boldsymbol{n}} \, dA = \int_{\mathbb{V}} \boldsymbol{\nabla} \cdot (\rho b \boldsymbol{V}) \, d\mathbb{V}, \qquad \int_{A} \dot{\boldsymbol{q}}'' \cdot \hat{\boldsymbol{n}} \, dA = \int_{\mathbb{V}} \boldsymbol{\nabla} \cdot \dot{\boldsymbol{q}}'' \, d\mathbb{V}$$

Localization Theorem

The *localization theorem* allows us to interchange integral forms of the conservation laws with their differential forms. The differential form is typically more practical for solving, especially with numerical methods. For a general function $f(x) = dF(x)/dx$ that is continuous and differentiable everywhere in the domain $\mathcal{D}$, if

$$\int_{\mathcal{D}} \left(\frac{dF}{dx}\right) dx = 0 \qquad \text{Integral form}$$

then,

$$\frac{dF}{dx} = 0 \qquad \text{Differential form}$$

everywhere in the domain. This can be restated for multiple variables!

Conservation of Mass (COM)

The general statement of conservation of mass is that the total change in mass of a system is zero. Writing this with the *material derivative*,

$$\frac{Dm}{Dt} = 0$$

Applying *Reynolds transport theorem* and rearranging[1],

$$\boxed{\frac{\partial}{\partial t}\int_{\mathbb{V}} \rho \, d\mathbb{V} = -\int_{A} \rho \boldsymbol{V} \cdot \hat{\boldsymbol{n}} \, dA} \qquad \text{ICOM II}$$

Applying *divergence theorem*,

$$\boxed{\frac{\partial}{\partial t}\int_{\mathbb{V}} \rho \, d\mathbb{V} = -\int_{\mathbb{V}} \boldsymbol{\nabla} \cdot (\rho \boldsymbol{V}) \, d\mathbb{V}} \qquad \text{ICOM II}$$

Applying *localization theorem* and rearranging,

$$\boxed{\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{V}) = 0} \qquad \text{DCOM}$$

If we define mass flowrates as $\dot{m} \equiv \int_A \rho \boldsymbol{V} \cdot \hat{\boldsymbol{n}} \, dA$, then COM Integral Form I simplifies to the form you may be used to in fluids and thermodynamics

$$\frac{\partial m_{\text{cv}}}{\partial t} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}}$$

[1] Here, "ICOM" means integral form of conservation of mass. Similarly, "DCOM" is the differential form.

Balance of Linear Momentum (BLM)

The general statement of balance of linear momentum is that the total change in momentum of a system is equal to the net force acting upon it. This is Newton's 2nd law of motion. Writing this with the *material derivative*,

$$\frac{D(m\boldsymbol{V})}{Dt} = \boldsymbol{F}_{\text{net}}$$

Applying *Reynolds transport theorem*,

$$\frac{\partial}{\partial t}\int_{\mathbb{V}} \rho\boldsymbol{V}\, d\mathbb{V} + \int_A \boldsymbol{V}\rho\boldsymbol{V}\cdot\hat{\boldsymbol{n}}\, dA = \boldsymbol{F}_{\text{net}}$$

The net force may be due to surface forces (tractions) or body forces (e.g., gravity). This may be written as[2],

$$\boxed{\frac{\partial}{\partial t}\int_{\mathbb{V}} \rho\boldsymbol{V}\, d\mathbb{V} + \int_A \boldsymbol{V}\rho\boldsymbol{V}\cdot\hat{\boldsymbol{n}}\, dA = \int_A \boldsymbol{\sigma}\cdot\hat{\boldsymbol{n}}\, dA + \int_{\mathbb{V}} \rho\boldsymbol{f}_b\, d\mathbb{V}} \qquad \text{IBLM I}$$

Applying *divergence theorem*,

$$\boxed{\frac{\partial}{\partial t}\int_{\mathbb{V}} \rho\boldsymbol{V}\, d\mathbb{V} + \int_{\mathbb{V}} \boldsymbol{\nabla}\cdot(\boldsymbol{V}\rho\boldsymbol{V})\, d\mathbb{V} = \int_{\mathbb{V}} \boldsymbol{\nabla}\cdot\boldsymbol{\sigma}\, d\mathbb{V} + \int_{\mathbb{V}} \rho\boldsymbol{f}_b\, d\mathbb{V}} \qquad \text{IBLM II}$$

Applying *localization theorem* and rearranging,

$$\boxed{\frac{\partial(\rho\boldsymbol{V})}{\partial t} + \boldsymbol{\nabla}\cdot(\boldsymbol{V}\rho\boldsymbol{V}) = \boldsymbol{\nabla}\cdot\boldsymbol{\sigma} + \rho\boldsymbol{f}_b} \qquad \text{DBLM I}$$

If we assume the fluid is a Newtonian fluid, the stress tensor $\boldsymbol{\sigma}$ can be written as

$$\boldsymbol{\sigma} = -p\boldsymbol{I} + \mu[\boldsymbol{\nabla V} + (\boldsymbol{\nabla V})^T]$$

where the first term describes axial pressures on the tensor main diagonal and the second term describes viscous shear on the tensor off diagonal. Further assuming gravity is the only body force, DBLM I simplifies to the form you may be used to in fluids, called the *Navier-Stokes equations*.

$$\boxed{\frac{\partial(\rho\boldsymbol{V})}{\partial t} + \boldsymbol{\nabla}\cdot(\boldsymbol{V}\rho\boldsymbol{V}) = -\boldsymbol{\nabla}p + \mu\boldsymbol{\nabla}^2\boldsymbol{V} + \rho\boldsymbol{g}} \qquad \text{DBLM II}$$

[2] Here, "IBLM" means integral form of conservation of momentum. Similarly, "DBLM" is the differential form.

Conservation of Energy (FLT)

The general statement of conservation of energy is that the total change in energy of a system is equal to the net heat transfer minus the net work acting upon it. This is the 1st law of thermodynamics. Writing this with the *material derivative*,

$$\frac{DE}{Dt} = \dot{Q}_{\text{net}} - \dot{W}_{\text{net}}$$

Applying *Reynolds transport theorem*,

$$\frac{\partial}{\partial t}\int_{\mathbb{V}} \rho e \, d\mathbb{V} + \int_A \rho e \boldsymbol{V} \cdot \hat{\boldsymbol{n}} \, dA = \dot{Q}_{\text{net}} - \dot{W}_{\text{net}}$$

The net heat transfer may be due to surface heat fluxes or internal heat generation. The net work may be due to pressure and viscous work (coming from $\boldsymbol{\sigma}$) or applied shaft work. This may be written as,

$$\boxed{\frac{\partial}{\partial t}\int_{\mathbb{V}} \rho e \, d\mathbb{V} + \int_A \rho e \boldsymbol{V} \cdot \hat{\boldsymbol{n}} \, dA = -\int_A \dot{\boldsymbol{q}}'' \cdot \hat{\boldsymbol{n}} \, dA + \int_{\mathbb{V}} \dot{q}'''_{\text{gen}} \, d\mathbb{V} - \dot{W}_{\text{p}} - \dot{W}_{\text{s}} - \dot{W}_{\text{v}}} \qquad \text{IFLT I}$$

The intensive energy e is the sum of internal energy, kinetic energy, potential energy, and so on. Adding the intensive pressure work $\dot{w}_p = p/\rho$ to the left-hand side allows us to write the flow energy in the area integral.

$$e = u + \frac{V^2}{2} + gz + \cdots, \qquad e_{\text{f}} = u + \frac{p}{\rho} + \frac{V^2}{2} + gz + \cdots = h + \frac{V^2}{2} + gz + \cdots$$

Applying *divergence theorem*,

$$\boxed{\frac{\partial}{\partial t}\int_{\mathbb{V}} \rho e \, d\mathbb{V} + \int_{\mathbb{V}} \boldsymbol{\nabla} \cdot (\rho e_{\text{f}} \boldsymbol{V}) \, d\mathbb{V} = -\int_{\mathbb{V}} \boldsymbol{\nabla} \cdot \dot{\boldsymbol{q}}'' \, d\mathbb{V} + \int_{\mathbb{V}} \dot{q}'''_{\text{gen}} \, d\mathbb{V} - \dot{W}_{\text{s}} - \dot{W}_{\text{v}}} \qquad \text{IFLT II}$$

Applying *localization theorem* and rearranging,

$$\boxed{\frac{\partial(\rho e)}{\partial t} + \boldsymbol{\nabla} \cdot (\rho e_{\text{f}} \boldsymbol{V}) = -\boldsymbol{\nabla} \cdot \dot{\boldsymbol{q}}'' + \dot{q}'''_{\text{gen}} - \dot{w}_{\text{s}} - \dot{w}_{\text{v}}} \qquad \text{DFLT I}$$

We typically only consider internal energy and neglect viscous and shaft work. Finally, we assume constant specific heats. Then, substituting *Fourier's Law* for the heat flux,

$$\boxed{\frac{\partial(\rho c_v T)}{\partial t} + \boldsymbol{\nabla} \cdot \left(\rho c_p T \boldsymbol{V}\right) = k \boldsymbol{\nabla}^2 T + \dot{q}'''_{\text{gen}}} \qquad \text{DFLT II}$$

Summary
For convection problems, these equations are coupled and must be solved together. Combining,

$$\boxed{\begin{gathered}\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{V}) = 0 \\ \frac{\partial (\rho \boldsymbol{V})}{\partial t} + \boldsymbol{\nabla} \cdot (\boldsymbol{V} \rho \boldsymbol{V}) = -\boldsymbol{\nabla} p + \mu \boldsymbol{\nabla}^2 \boldsymbol{V} + \rho \boldsymbol{g} \\ \frac{\partial (\rho c_v T)}{\partial t} + \boldsymbol{\nabla} \cdot \left(\rho c_p T \boldsymbol{V}\right) = k \boldsymbol{\nabla}^2 T + \dot{q}'''_{\text{gen}}\end{gathered}}$$

Under the further assumptions of constant and uniform properties, incompressible, and negligible pressure effects, these reduce to the *convective heat transfer equations*

$$\boxed{\begin{gathered}\boldsymbol{\nabla} \cdot \boldsymbol{V} = 0 \\ \rho \frac{\partial \boldsymbol{V}}{\partial t} + \rho (\boldsymbol{V} \cdot \boldsymbol{\nabla}) \cdot \boldsymbol{V} = -\boldsymbol{\nabla} p + \mu \boldsymbol{\nabla}^2 \boldsymbol{V} + \rho \boldsymbol{g} \\ \rho c_p \left(\frac{\partial T}{\partial t} + \boldsymbol{\nabla} \cdot (T \boldsymbol{V}) \right) = k \boldsymbol{\nabla}^2 T + \dot{q}'''_{\text{gen}}\end{gathered}} \qquad \text{CHT}$$

Finally, if we consider a stagnant fluid, the first equation becomes degenerate[3], and we recover the *heat conduction equation* and *hydrostatic equation*.

$$\boxed{\begin{gathered}\boldsymbol{\nabla} p = \rho \boldsymbol{g} \\ \rho c_p \frac{\partial T}{\partial t} = k \boldsymbol{\nabla}^2 T + \dot{q}'''_{\text{gen}}\end{gathered}} \qquad \text{HCE}$$

[3] Basically, we get $0 = 0$ and conservation of mass doesn't tell us anything.